

HSH HEAVY SAFE HELICOPTER

HSH-00 | REVISION 22 FULL

FINAL GENERATION 1 MASTER ENGINEERING BASELINE

Complete Content | No Deletion | No Blank Pages | Uniform Portrait

Founder: Amirbahador Atai | **Document:** HSH-00 | **Baseline:** complete Rev20 technical content retained | **Enhancement:** §§32–41 diligence layer

CONTROL PRINCIPLE: This package preserves the complete Rev20 baseline content and adds ten engineering-control sections (§§32–41). No Rev20 technical value, requirement, open item, table or claim is intentionally removed. Composition is continuous so that near-empty white pages are not produced.

DOCUMENT USE: Single portrait A4 master baseline for data-room evaluation. TARGET, ASSUMPTION, STUDY, CALC and OPEN values are not demonstrated performance. Not a certification approval, not a filed patent, not proof of achieved survival probability.

DOCUMENT CONTROL

Field	Value
Document Title	HSH Heavy Safe Helicopter — Generation 1 Master Engineering Baseline
Document Number	HSH-00
Revision	Rev 22 FULL
Supersedes	HSH-00 Rev 12 through Rev 21 presentation packages
Status	FINAL G1 CONCEPTUAL MASTER BASELINE
Founder	Amirbahador Atai
Development Stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Classification	Proprietary Concept Development Material
Certification note	Not a certification approval
Patent note	Not a filed patent

***Engineering Integrity Statement:** TARGET, ASSUMPTION, STUDY, CALC and OPEN values are not demonstrated performance. Critical claims require analytical closure, simulation, ground testing, integrated testing and flight-test evidence. Every status label shall be interpreted as defined in §5.*

REVISION HISTORY

Rev	Primary Change	Status
Rev 12	26,000 kg / 26.0 m / 4.0 m baseline; 8-canopy recovery; single 6 s target; 10–12 occupants	Superseded
Rev 13	Main-rotor separation removed from core; staged S0–S5; 3+2 canopies; Risk Register; Prior Art	Superseded
Rev 14–16	Full narrative depth; helium-hybrid trade posture; competitive IP narrowing	Superseded
Rev 17–18	HSH identity unified; 8-person lock; commercial content separated; Maturity Framework	Superseded
Rev 19	16-airbag architecture; seat specification; OI-16/OI-17 registered	Superseded
Rev 20	Seat stroke ≥150 mm / ≤6,670 N; airbag mass 200 kg; OI-18; IP-D; TRL; competitive; defined-vs-open; geometry; mission	Superseded as presentation base
Rev 21–22	Diligence §§32–41; portrait uniformity; no-blank dense composition of full content	CURRENT FULL

TOP-LEVEL TECHNICAL BASELINE — MASTER TABLE

Parameter	Rev 22 Value	Status	Key Note
Design mass	26,000 kg	[TARGET]	Planning closure; bottom-up pending
Maximum capsule occupancy	8 persons	[LOCKED LIMIT]	Generation 1 hard limit
Main rotor diameter	26.0 m	[TARGET]	Benchmark comparison §4.1
Main rotor disk area	530.93 m²	[CALC]	$\pi \times (13.0)^2$
Disk loading	48.97 kg/m²	[CALC]	See §4.1
Tail rotor diameter	4.0 m	[TARGET]	Anti-torque closure pending
Tail rotor disk area	12.57 m²	[CALC]	$\pi \times (2.0)^2$
Capsule mass	7,000 kg	[TARGET]	Bottom-up pending Doc 113
Range target	1,000 km	[TARGET]	Mission model pending §21
Ideal hover power	≈ 3.57 MW	[CALC]	Actuator disk, ISA SL
Shaft power @ FM 0.75	≈ 4.76 MW	[CALC]	Study value only
Propulsion	Two turboshaft engines	[REQ/STUDY]	Engine selection open
Overall length	16.5–18.0 m	[TARGET/STUDY]	Layout reconciliation pending

Parameter	Rev 22 Value	Status	Key Note
Fuselage width (max)	4.0–4.5 m	[TARGET/STUDY]	
Overall height	4.8–5.3 m	[TARGET/STUDY]	
Main rotor separation	Not part of core safety case	[DEFERRED — RESEARCH TRACK]	\$12
Recovery canopies	3 primary + 2 reserve	[STUDY]	\$11
Separation timeline	Staged S0–S5	[TARGET/OPEN per stage]	\$13
Occupant airbags	16 total (2 per occupant)	[TARGET/STUDY]	\$10A; OI-18
Crashworthy seat stroke	≥150 mm dynamic stroke	[TARGET/STUDY]	\$10B
Peak spinal load limit	≤6,670 N reference criterion	[TARGET/STUDY]	FAR 27.562 proxy; \$10B
Development gate	G0 approaching closure	[CURRENT]	G1–G8 not started

MASS ACCOUNTING — 26,000 kg

Mass Group	Planning Mass	Status
Survival capsule, complete	7,000 kg	TARGET
Main airframe excluding capsule	5,000 kg	ASSUMPTION / STUDY
Main rotor, hub and controls	2,100 kg	ASSUMPTION / STUDY
Tail rotor, gearbox and drive	480 kg	ASSUMPTION / STUDY
Engines, transmission, gearboxes and mounts	3,100 kg	ASSUMPTION / STUDY
Fuel system + full usable fuel	3,400 kg	ASSUMPTION / STUDY
Rapid separation system	1,000 kg	ASSUMPTION / STUDY
Avionics, electrical, hydraulic, thermal, control	1,600 kg	ASSUMPTION / STUDY
Operating fluids and consumables	320 kg	ASSUMPTION / STUDY
Growth / configuration reserve	2,000 kg	PROGRAM RESERVE
TOTAL	26,000 kg	CALC / ACCOUNTING CLOSURE

CAPSULE COMPONENT MASS — 7,000 kg

Capsule Component	Planning Mass	Notes
Primary crashworthy cell / internal roll-cage	1,950 kg	CFRP + titanium nodes
Outer capsule shell / fairing / thermal protection	500 kg	
Eight fixed landing supports + progressive energy absorbers	850 kg	
Eight crashworthy seats + restraints (≥150 mm stroke, ≤6,670 N)	700 kg	Rev20 corrected spec
Sixteen occupant airbags + inflators + sensors + wiring	200 kg	Rev20 corrected (was 80–120 kg)
Emergency exits / doors, frames and seals	250 kg	
Distributed multi-canopy recovery system (3 primary + 2 reserve)	650 kg	
Independent capsule emergency power + batteries	250 kg	
Life support / emergency oxygen / environmental	200 kg	
Avionics / sensing / communications / locator / lighting	250 kg	
Separation interfaces + capsule-side release hardware	350 kg	
Internal secondary structure / floor / partitions / padding	300 kg	Reduced to absorb airbag mass
Fire / smoke detection and suppression	150 kg	
Water landing / flotation / sealing	150 kg	
Inspection / access / replaceable protection	100 kg	
Capsule configuration growth allowance	150 kg	
TOTAL	7,000 kg	CALC / CLOSURE

Rev20 mass note retained: Airbag system mass corrected from 80–120 kg to **200 kg** based on automotive-class benchmarks scaled to 16-bag aviation-grade system with ECU, pyrotechnic harness, mounting and qualification margin. Corresponding reduction taken from internal secondary structure. Total capsule budget unchanged at 7,000 kg. If detailed design exceeds 200 kg, growth uses the 150 kg allowance; further growth triggers configuration-controlled impact assessment.

1. BASELINE FREEZE AND CONTROLLED IDENTITY

1.1 Programme Identity

The programme name is **HSH Heavy Safe Helicopter**. The controlling document is **HSH-00 Revision 22 FULL**, Founder: Amirbahador Atai. All prior references to HSC-00 or earlier HSH-00 revisions are superseded for configuration control. Rev22 FULL carries the complete Rev20 technical baseline plus the diligence enhancement layer.

1.2 Occupancy Lock

8 persons maximum [LOCKED LIMIT]. This is a Generation 1 hard configuration constraint. Any future change requires formal configuration control and re-analysis of geometry, mass, CG, inertia, seating, restraints, airbag count, life support, egress, structural loads, recovery sizing and certification basis.

1.3 What Rev20/22 Changed Relative to Rev19

Change	Rev 19	Rev 20/22	Rationale
Seat specification	≥50 cm ergonomic foam	≥150 mm dynamic stroke; ≤6,670 N spinal load	Foam thickness is not an aviation crashworthy-seat criterion
Airbag mass budget	80–120 kg	200 kg	Automotive benchmark scaled to 16 aviation-grade units
Sealed-capsule airbag hazard	Not explicitly registered	OI-18 added	16 airbags in sealed volume: overpressure and gas-toxicity risk
IP-D claim	Not present	Added	System-level combination claim stronger than any single IP-A/B/C
TRL / competitive / defined-vs-open / geometry / mission	Not present	Added	Required for serious technical due-diligence

1.4 What Is Retained Unchanged

Every requirement, status label, open item OI-01 through OI-17 (plus OI-18), risk register entry, verification gate, interface definition, safety layer, Safety Kernel precondition set, staged separation timeline, recovery architecture, prior art posture, and engineering maturity framework from Rev18/Rev19/Rev20 is retained. Only presentation defects (blank pages, mixed orientation, empty cover boxes) are corrected in Rev22 FULL.

2. PURPOSE AND SCOPE

2.1 Programme Objective

HSH Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain — not reliance on any single device.

2.2 Core Scoping Decision

The survival chain is designed to work assuming the main rotor is never separated from the aircraft. This deliberate decision was introduced in Rev13 and is retained. No credible trajectory, gyroscopic, or structural analysis yet exists to support main-rotor separation from a descending occupied capsule. Building the core safety case on an unproven capability would undermine confidence in the entire architecture and in all primary IP claims.

2.3 Scope Coverage

This document covers: the aircraft; the occupied survival capsule; the rapid selective separation system (non-rotor components only); safety initiation paths; deterministic safety authorization; stabilization; recovery; impact attenuation; post-event survival; and verification interfaces. It does not cover detailed structural analysis results, certified engine data, qualification test results, airworthiness findings, or IP legal opinions — those are evidence-gate deliverables.

2.4 Design Philosophy

Separation is a controlled emergency mode, authorized only when defined conditions are satisfied. The survival architecture supports multiple outcomes in priority order: (1) conventional flight recovery; (2) autorotation; (3) selective hazard/mass reduction; (4) capsule separation and parachute recovery; (5) energy-absorbing landing; (6) post-landing rescue. The most extreme mode shall not be required for manageable failures.

2.5 Human Authority

Human emergency authority is preserved and non-bypassable. No AI system, optimization algorithm, or predictive model shall override the crew's ability to command, inhibit, or terminate an automatic survival sequence. This is a design requirement, not a guideline.

3. WHAT IS DEFINED TODAY vs WHAT IS OPEN

This section is the first thing a technical due-diligence team should read.

Domain	What is Defined	What Remains Open	Gate
System architecture	Six-layer survival; Safety Kernel; state machine; staged separation; 3+2 recovery; 8-support landing	All engineering analysis	G1
Mass accounting	26,000 kg and 7,000 kg planning models; component-level allocation	Bottom-up verification; uncertainty and growth	G1
Capsule structure	CFRP roll-cage concept; titanium nodes; load path description	FEA; crash load analysis; fatigue	G5
Separation system	S0–S5 timeline; candidate non-rotor components; mandatory protections	S3 mechanism; 6-DOF; ground demonstration	G2–G3
Recovery system	3+2 canopy architecture; sizing rationale; failure tolerance concept	Canopy sizing; inflation dynamics; min deployment altitude	G6

Domain	What is Defined	What Remains Open	Gate
Landing supports	8-point concept; energy absorber; asymmetric-failure requirement	FEA; drop tests; terrain interaction	G5
Crashworthy seats	≥150 mm stroke; ≤6,670 N criterion; 8 positions	Seat design; foam/stroking-leg; dynamic tests	G2–G5
Airbag system	16-bag concept; 2 per occupant; placement; 200 kg budget	Packaging; deployment logic; sealed-volume gas (OI-18); sled tests	G2–G5
Safety Kernel	Precondition set defined; verification requirements stated	Software/hardware implementation; formal verification	G1–G2
IP claims	IP-A, IP-B, IP-C, IP-D conceptually defined; prior art posture stated	Professional prior-art search; claim drafting; filing	External legal
Certification basis	CS-29 referenced	Formal regulatory engagement; means of compliance	External regulatory
Performance	Hover power study (3.57 MW); disk loading benchmark	All performance analysis; cruise; OEI; range closure	G1
Mission profile	1,000 km range target; 8 occupants	Fuel model; cruise speed; altitude; reserve policy	G1

4. AIRCRAFT GEOMETRY AND CONFIGURATION

4.1 Rotor Disk Loading — Benchmark Comparison

Aircraft	MTOW	Main Rotor Diameter	Disk Loading
Mi-26 (Russia — heaviest production)	56,000 kg	32.0 m	~55 kg/m ²
CH-53K King Stallion (US heavy-lift)	38,400 kg	24.1 m	~58–60 kg/m ²
CH-47F Chinook (tandem, per disk)	22,680 kg	18.3 m per rotor	~43 kg/m ² per disk
Sikorsky S-92 (medium offshore)	12,020 kg	17.17 m	~52 kg/m ²
AW139 (medium — market neighbor)	6,800 kg	13.8 m	~45 kg/m ²
HSH Gen1 Rev22	26,000 kg	26.0 m	48.97 kg/m ²

Design rationale: HSH's disk loading is deliberately lower than Mi-26 and CH-53K. This trades rotor size for: (a) greater hover power margin — needed given the additional mass of the capsule, separation system, airbags and crashworthy seats; (b) reduced downwash velocity at the eight fixed landing supports; and (c) lower tip-speed requirement, reducing rotor noise. Confirmed or revised at Document 063 closure.

4.2 Aircraft Geometry Summary

Geometry Item	Preliminary Definition	Status
Overall length	Approx. 16.5–18.0 m	[TARGET/STUDY]
Maximum fuselage width	Approx. 4.0–4.5 m	[TARGET/STUDY]
Overall height	Approx. 4.8–5.3 m	[TARGET/STUDY]
Main rotor diameter	26.0 m	[TARGET]
Tail rotor diameter	4.0 m	[TARGET]
Rotor arrangement	Single main rotor + tail rotor	[REQ/STUDY]
Landing supports	8 fixed, short, mechanical supports under capsule	[TARGET]
Retractable landing gear	None	[LOCKED CONCEPT]
Capsule position	Central, integrated within primary airframe	[LOCKED CONCEPT]
Capsule normal-flight load path	Primary attachment system with controlled emergency release	[REQ]
Capsule emergency load path	Independent final landing element after verified clearance	[REQ]

5. BASELINE STATUS AND CONFIGURATION CONTROL

5.1 Status Label Definitions

Status Label	Precise Meaning
[LOCKED CONCEPT]	Retained as baseline design principle; not to be changed without formal impact assessment
[LOCKED LIMIT]	Hard Generation 1 constraint; requires formal change process and full re-analysis to alter
[REQ]	Top-level requirement; binding on the design unless formally waived
[TARGET]	Numerical or functional development objective; not demonstrated
[ASSUMPTION / STUDY]	Working engineering assumption for preliminary sizing; to be replaced by analysis or test
[CALC]	Calculated from stated assumptions; inputs are themselves assumptions or targets
[OPEN]	Requires analysis, simulation, test, supplier data or design decision
[OPEN — CRITICAL]	Open item with material effect on the core safety case; priority closure required
[DEFERRED — RESEARCH TRACK]	Excluded from licensable core baseline; not to be presented as core product
[TBD — DEPENDENCY]	Cannot be fixed until a dependent activity closes
[PRIOR ART STATUS: NOT YET SEARCHED]	No novelty conclusion implied; professional search required

5.2 Configuration Change Triggers

A formal configuration change is required for any of the following: MTOW; main rotor diameter; tail rotor diameter; capsule mass; occupant count; separation sequence architecture; parachute architecture; landing-support count or geometry; airbag count or placement; seat stroke specification; Safety Kernel precondition set; any safety-layer definition. Each change triggers an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

6. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

The six layers are **functionally distinct** and **independently capable** — not merely redundant. Independence means physically, electrically, and software-separate implementations that cannot both be defeated by any single common-cause failure.

Layer	Function	Primary Purpose	Independence Requirement
L1	Primary automatic emergency detection and initiation	Detect catastrophic states; initiate validated survival sequence when Safety Kernel authorizes	Separate processor, sensor suite, power bus from L2
L2	Independent backup initiation	Second segregated automatic path if L1 unavailable or in disagreement	Different physical location, software architecture, power source from L1
L3	Independent manual/crew authority	Crew initiation, inhibit and emergency authority without L1/L2	Hardwired path bypassing L1/L2; software cannot disable
L4	Rapid selective physical separation	Release designated non-rotor hazards using controlled mechanisms	Dual independent actuation paths; no single-point failure defeats both
L5	Direct-impact survival	Protect occupants via capsule structure, restraints, seats and eight energy-absorbing supports	Passive structural system; no active components required for primary function
L6	Recovery and post-event survival	Deploy recovery canopies; locate occupants; support rescue	Independent emergency power; independent comms; independent locator beacon

AI Boundary — Non-Negotiable Principle: AI or advanced prediction may assist detection, classification and decision support. **Deterministic safety logic authorizes safety-critical actuation. AI confidence is never equivalent to safety authorization.** This principle is non-negotiable from a certification perspective and is the engineering basis for IP-A (§17).

Survival Chain: PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT (8 supports + crashworthy seats + 16 airbags) → SURVIVE / LOCATE / RESCUE → AUDIT

7. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

The Safety Kernel prevents AI, digital-twin models, and predictive systems from directly commanding hazardous separation without satisfying a defined set of physically verifiable preconditions. It is the boundary between the decision layer and the actuation layer.

7.2 Safety Kernel Precondition Set

Precondition Category	Required Evaluation
Altitude and airspeed	Above minimum recovery altitude; within separation dynamic-pressure envelope; within parachute deployment envelope accounting for sequence time
Attitude and angular rates	Pitch, roll, yaw within acceptable separation envelope; angular rates within acceptable separation rate limits
Rotor and engine state	Rotor RPM documented; engine/fire/fuel hazard state known and consistent with separation sequence
Structural and attachment	Capsule attachment health confirms release mechanisms functional; structural health consistent with planned separation
Trajectory and clearance	Acceptable clearance between released components and capsule/parachutes confirmed (real-time or pre-computed envelope)
Recovery system	Parachute deployment envelope satisfiable after separation; recovery integrity confirmed
Power and communication	Independent survival power available; communication loss does not inhibit Safety Kernel
Human authority	Manual crew inhibit confirmed; rollback/fallback availability assessed
Airbag authorization	Impact state valid; deployment permitted under deterministic preconditions; capsule-interior environment suitable

7.3 Kernel Decision Table

Decision	Minimum Design Question	Verification Method
Authorize separation	Are all mandatory preconditions simultaneously true?	Formal logic review + fault injection
Inhibit separation	Can an unsafe state block actuation even under software fault?	Fault injection + power-loss test
Fallback mode	What happens if a sensor or command path is lost?	Failure-mode testing per FMEA
Human override	Is authority bounded, deterministic and physically independent?	HMI + system integration test
Airbag authorize	Is impact state valid and deployment safe in closed capsule?	Logic review + sled/impact test + gas analysis
Audit record	Can every emergency event be fully reconstructed?	Data-recording verification test

The Safety Kernel shall be implemented with: formally verified software or hardware logic; documented requirement set with code traceability; independence from AI/ML layers; segregated power supply; independent communication path to actuation hardware; protection against command injection; and an immutable audit record. No update or in-flight reprogramming shall modify the Safety Kernel precondition set without a formal change process and re-verification.

8. CONTROLLED RAPID SEPARATION SYSTEM

The system is selective, staged, and controlled — not a blanket jettison of everything. State-dependent release logic means the set of components released in a fire emergency differs from the set in a structural emergency. Every component designated as separable must be confirmed through structural, aerodynamic, dynamic, fire, control, trajectory and collision analysis.

8.2 Candidate Separable Components

Component	Condition for Separation	Analysis Requirement
Main engines (both)	After fuel isolation; where structurally separable	Trajectory, structural, thermal, fire analysis
Selected transmission/drive components	Where release is structurally and dynamically acceptable	6-DOF dynamics, CG shift analysis
Tail rotor assembly + gearbox + drive	Where aerodynamically and dynamically safe	Anti-torque state, trajectory, collision analysis
Primary fuel tanks + major fuel lines	After confirmed fuel isolation	Fire hazard analysis, isolation timing
Selected external structural panels	High-mass, not on capsule primary load path	Structural, trajectory, collision analysis
Main rotor blades and hub	EXCLUDED from core candidate list	Deferred research track — §12

8.3 Mandatory Release Protections

Protection	Requirement
Positive mechanical retention	Active positive lock during all normal flight; release requires deliberate actuation
Electrical/logic inhibit	Release electrically inhibited in normal flight state; specific arming sequence required
Independent command paths	Primary and backup paths physically and electrically independent
Defined arm/fire sequence	Documented, tested, protected against out-of-sequence commands
Release-state sensing	Positive confirmation of attached/released state at each stage
Fuel/energy isolation	Isolation precedes or is concurrent with release; no new uncontrolled hazard created
Asymmetric/partial-release detection	Full reporting of any case where less than the designated set is released
Post-release monitoring	Track released components where practically possible
Capsule-clear verification	Capsule release not authorized until all non-rotor components confirmed clear
Failed-release fallback	Safe state defined for every possible failed-release configuration

8.4 Separation System Mass Budget — 1,000 kg

Separation System Component	Mass
Primary capsule release mechanisms and locks	240 kg
Independent backup release mechanisms	150 kg
Local structural reinforcement and release fittings	160 kg
Independent initiators / actuators	90 kg
Release-state sensors and monitoring	65 kg
Independent power / signalling and protected wiring	70 kg
Fuel / energy isolation interfaces	55 kg
Retention, inhibit and safety-interlock hardware	50 kg
Configuration growth allowance + integration margin	120 kg
TOTAL	1,000 kg

9. OCCUPIED SURVIVAL CAPSULE

The capsule is not a cabin with a parachute. It is a structurally independent survival system that functions as the passenger compartment during normal flight and becomes the final landing boundary in a catastrophic emergency. Every subsystem inside the capsule must function independently of the main airframe after separation. Primary structure: continuous CFRP roll-cage with metallic/titanium nodes at load-introduction points. CFRP provides high specific strength; titanium nodes provide fatigue-resistant, corrosion-resistant, impact-resistant interfaces at maximum stress concentration points. Continuous roll-cage geometry provides controlled deformation paths — progressive outward deformation rather than inward collapse.

Capsule Parameter	Rev 22 Definition	Status
Occupancy	8 persons maximum	[LOCKED LIMIT]
Complete capsule mass	7,000 kg planning target	[TARGET]
Primary structure	Continuous CFRP roll-cage, titanium nodes	[CONCEPT]
Seat type	Energy-absorbing, ≥150 mm stroke, ≤6,670 N peak spinal load	[TARGET/STUDY]
Airbag system	16 total (2 per occupant)	[TARGET/STUDY]
Landing supports	8 fixed, progressive energy absorbers	[TARGET]
Recovery canopy architecture	3 primary + 2 reserve	[STUDY]
Independent emergency power	Battery-based, independent of airframe bus	[TARGET]
Life support	Emergency oxygen, minimum post-landing endurance	[TARGET]

Capsule Parameter	Rev 22 Definition	Status
Capsule-clear verification	Required before recovery deployment	[REQ]
Post-landing stability	8-point architecture; asymmetric/failed-leg analysis required	[REQ]
Water landing	Flotation, sealing for ditching scenario	[TARGET]
Emergency egress	Multiple exits rated for crash loads; airbag-compatible paths	[TARGET]

10. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

Eight short, thick, fixed mechanical supports are integrated with the capsule lower structure. They are not conventional helicopter landing gear. Key rationale: **Capsule-integrated** — supports remain with occupants after separation; **Redundancy** — loss of one or two supports still leaves six or seven functional; **Energy absorption** — each support incorporates a progressive mechanical energy-absorber with controlled, bounded crush force throughout deformation without a hard metallic stop.

Impact Velocity	Total KE (7,000 kg capsule)	KE per Support (÷8)
6 m/s	126.0 kJ	15.75 kJ
8 m/s	224.0 kJ	28.0 kJ
10 m/s	350.0 kJ	43.75 kJ

These are sizing illustrations, not allowable impact energies. Actual survival depends on pulse shape, attitude, angular rates, terrain and structural response. Seat stroke and airbags provide additional occupant load management after the supports arrest capsule motion.

Case	Analysis Focus	Required Evidence
All 8 supports, flat terrain	Load sharing and stability	FEA + drop test
1 support failed	Redistribution and local overload	FEA + test
2–3 supports failed	Survivability boundary	Sensitivity study
Sloped terrain (<15°)	Roll/tip tendency, asymmetric load transfer	Terrain simulation
Soft ground / snow	Penetration and energy dissipation	Ground interaction test
Water impact (ditching)	Hull sealing and flotation	Water impact test
Oblique combined loads	Lateral + vertical + longitudinal	6-DOF dynamic simulation

10A. SIXTEEN-AIRBAG OCCUPANT PROTECTION ARCHITECTURE

Item	Rev 22 Baseline
Occupant positions	8 (locked maximum)
Airbags per occupant	2
Total airbags	16
Primary set	8 — one per seat, head/torso / principal impact direction
Secondary set	8 — one per seat, lateral / lower-body / secondary direction
Status	[TARGET/STUDY]
Mass planning (corrected)	≈ 200 kg (16 aviation-grade modules + inflators + ECU + pyrotechnic harness + mounting)
Verification	Sled tests, inflation tests, interference checks, gas-toxicity review, egress compatibility

Placement principles: Eight primary + eight secondary; shall not obstruct egress; compatible with CFRP roll-cage, seats and restraints; deployment only under deterministic Safety Kernel preconditions; airbag authorization passes through Safety Kernel — not an independent system that can deploy outside the authorization chain.

Mass correction rationale: Rev19 estimated 80–120 kg. Aviation-grade modules at 1.5–2.0 kg each × 16 ≈ 24–32 kg module mass. Adding ECU (8–15 kg), pyrotechnic harness (20–30 kg), mounting (30–40 kg), wiring (15–20 kg), and qualification margin yields approximately 150–220 kg, rounded to **200 kg**. The Rev19 estimate was insufficient and would have created a false closure of the capsule mass budget.

Sealed-Capsule Deployment Hazard — OI-18 (CRITICAL): Sixteen airbags deploying simultaneously in a sealed capsule of limited internal volume create risks not present in automotive applications: (a) overpressure — rapid gas generation may transiently exceed safe ear-drum and sinus pressure limits; (b) gas composition — propellant gases dilute capsule oxygen; (c) temperature — inflator heat in a sealed volume; (d) particulate — sodium compounds from certain propellant chemistries. Mitigation options include staggered sequential deployment, vented inflators, or reduced inflator energy for the closed-capsule environment. Must be analyzed before deployment logic is locked.

10B. EIGHT CRASHWORTHY SEATS — CORRECTED SPECIFICATION

Rev19 specified “≥50 cm ergonomic foam seating-surface thickness.” This is not a recognized aviation crashworthy-seat criterion. FAR 27.562 (and CS-29 equivalent) evaluates seats by occupant injury criteria, not foam thickness. Soft foam can compress fully under crash loads in milliseconds, providing minimal protection after initial compliance. The correct engineering specification is a minimum dynamic stroke and a peak load limit.

Item	Rev 22 Baseline
Number of seats	8
Minimum dynamic stroke	≥ 150 mm (combined seat/absorber/floor deformation path)

Item	Rev 22 Baseline
Peak spinal compressive load limit	≤ 6,670 N (FAR 27.562 reference criterion, proxy)
Force-displacement response	Progressive — bounded crush force throughout stroke, no hard stop at stroke limit
Seat architecture	Energy-absorbing leg + foam/elastomer cushion combination [OPEN — detailed design]
Function	Limit peak loads transmitted to pelvis, spine and lower extremities during hard landing
Works with	8 landing supports + 16 airbags + 5-point restraints + roll-cage
Status	[TARGET/STUDY] — density curve, stroke profile, fire rating, mass, dynamic performance [OPEN]
Mass	Included in 700 kg seats + restraints capsule line
Verification	Dynamic seat tests per FAR 27.562 or CS-29 equivalent; combined drop/sled cases

11. RECOVERY ARCHITECTURE — 3 PRIMARY + 2 RESERVE CANOPIES

Three large primary canopies are preceded by crewed spacecraft recovery systems at comparable mass. Fewer larger canopies reduce simultaneous-inflation complexity and inter-canopy collision risk at low altitude — the failure mode most common in real rotorcraft emergencies.

Recovery Mode	Use Case	Status
Controlled flight recovery	Recoverable disturbance	[REQ/STUDY]
Autorotation	Engine/power loss when altitude/rotor state permit	[REQ/STUDY]
Selective mass/hazard separation	Fire, structural or combined emergency	[REQ/CONCEPT]
Capsule separation	Catastrophic state	[REQ/CONCEPT]
3 primary recovery canopies	Primary post-separation vertical recovery	[STUDY]
2 reserve canopies	On detected primary failure only	[STUDY]
8 energy-absorbing landing supports	Final ground impact attenuation	[REQ/TARGET]
Crashworthy seats (≥150 mm) + 16 airbags	Occupant load management after ground contact	[TARGET/STUDY]
Post-landing survival	Locator, comms, lighting, emergency power, rescue interface	[REQ]

For 7,000 kg capsule at 7.5 m/s target descent rate: **A_{total} ≈ 2,983 m²**; per canopy (÷3): **≈ 994 m²** each; corresponding diameter **≈ 35–36 m** per primary canopy. Consistent with spacecraft precedent (including Soyuz-class multi-main canopy recovery). Detailed sizing: Document 159. The minimum deployment altitude is **[OPEN — CRITICAL] (OI-02)**. Most rotorcraft accidents occur at low altitude.

12. CORE SAFETY CASE — REBUILT WITHOUT MAIN-ROTOR SEPARATION

The single largest credibility risk in Rev12 was the implicit dependency of the safety case on main-rotor separation — a capability for which no credible 6-DOF simulation, structural trajectory analysis, or physical evidence exists. A 26 m rotor carries several hundred megajoules of rotational kinetic energy and a gyroscopic moment resisting rapid attitude change. Removing this dependency from the core safety case is what makes the remaining architecture defensible.

The survival chain works assuming the main rotor is never separated.

1. Controlled flight recovery / autorotation — no separation required.
2. Selective non-rotor hazard separation — engines, fuel tanks (post-isolation), tail rotor/drive, fire-load panels.
3. Capsule release from airframe — capsule departs beneath/behind the still-attached rotor disk.
4. Capsule descent under 3+2 canopy system.
5. Impact attenuation via eight fixed supports + crashworthy seats + 16 airbags.
6. Post-landing survival — independent power, life support, locator, communications.

Main-rotor separation is retained as Document 080 deferred research, explicitly excluded from the licensable core baseline and from primary IP claims. If independent 6-DOF analysis and mechanism test evidence later demonstrate safe separation, it may be reintroduced through the configuration change process.

13. STAGED SEPARATION TIMELINE

Stage	Action	Target Window	Confidence Basis	Status
S0	Validate emergency state; arm path; isolate fuel/electrical/energy	0.0–0.5 s	Comparable to ejection-seat arming	[TARGET]
S1	Fuel isolation and designated fire-hazard separation	0.5–1.8 s	Comparable to crash fuel-shutoff	[TARGET]
S2	Engine + tail-rotor drive separation (non-main-rotor)	1.8–3.5 s	Comparable to cargo-release, scaled	[TARGET/OPEN]
S3	Capsule-to-airframe primary structure release	3.5–5.0 s	OPEN — CRITICAL: no direct certified analog	[OPEN — CRITICAL]
S4	Capsule clearance verification	5.0–5.8 s	Sensor-response-limited	[TARGET]
S5	Recovery canopy authorization	After S4, gated on clearance	Not time-gated; correctness first	[REQ]

If the architecture cannot safely achieve these windows, targets shall be revised under configuration control — not achieved by reducing verification rigor. Partial and asymmetric separation are first-class emergency states at every stage.

14. EMERGENCY STATE MACHINE

State	Entry Condition	Primary Action	Exit / Next
1 — NORMAL FLIGHT	All systems nominal	Continuous monitoring; locks engaged	Anomaly → 2
2 — ANOMALY DETECTED	Sensor/model flags abnormality	Increase monitoring; begin confirmation	Confirmed → 3; Rejected → 1
3 — CONFIRMATION	Potential condition detected	Independent evidence and validity checks	Confirmed → 4
4 — IMPACT ASSESSMENT	Condition confirmed	Estimate authority, altitude, airspeed, fire/structural risk	Complete → 5
5 — RECOVERY ATTEMPT	Assessment complete	Conventional control, OEI, autorotation	Succeeds → 1; Fails → 6
6 — SURVIVAL AUTHORIZATION	Recovery failed	Safety Kernel evaluates all preconditions	Authorized → 7; Not met → degraded
7 — STAGED SEPARATION	Authorized	Execute designated release set (non-rotor)	Complete → 8
8 — CAPSULE CLEARANCE	Separation executed	Verify clearance; identify residual attachments	Clear → 9; Partial → partial-release
9 — RECOVERY DEPLOYMENT	Clearance verified	Authorize parachute within validated envelope	Deployment confirmed → 10
10 — DESCENT MANAGEMENT	Canopies deployed	Monitor attitude, rates, canopy state, support readiness	Ground imminent → 11
11 — LANDING ATTENUATION	Ground contact	Supports + seats + airbags all load simultaneously	Landing complete → 12
12 — POST-LANDING SURVIVAL	Stabilized	Locator active; comms active; life support active	Rescue confirmed → 13
13 — AUDIT	Rescue complete	Preserve all event data, sensor states, command history	Data secured — END

15. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE CONTROL

Failure / Hazard	Severity	Required Design Response
Unintended capsule separation in normal flight	Catastrophic	Mechanical locks + electrical inhibits + command validation + monitoring + quantitative safety objective
Partial separation (incomplete component release)	Critical	State sensing + asymmetric-load analysis + defined degraded recovery mode
Asymmetric separation	Critical	6-DOF dynamic analysis + CG/inertia update + post-release control assessment
Release mechanism failure	Critical	Independent path + defined safe fallback; no assumption of perfect release
Sensor loss or bad data	Hazardous	Validity monitoring + redundant/diverse sensing + safe-state logic
L1 automatic detection failure	Hazardous	Independent L2 backup + L3 manual crew initiation
Power-bus failure	Critical	Independent survival power for critical functions
Communication loss	Hazardous	Local deterministic safety logic; communication not in authorization path
AI/ML model failure	Hazardous	Deterministic Safety Kernel remains authoritative
Fire during separation	Critical	Fuel isolation + fire containment + coordinated sequencing
Released component collision with capsule/parachutes	Catastrophic	Trajectory analysis + clearance envelopes + controlled sequencing
Parachute deployment outside envelope	Critical	Hard inhibits + validated deployment conditions
Landing-support failure (one or more)	Hazardous	8-support redundancy + asymmetric/failed-leg structural analysis
Airbag inadvertent deployment in flight	Critical	Deterministic inhibit logic + dual-path sensing + mechanical safing
Airbag non-deployment on impact	Hazardous	Backup initiation path + sled test verification
Airbag overpressure / gas toxicity in sealed capsule	Critical	OI-18: sealed-volume gas analysis required
Seat stroke bottom-out at extreme load	Hazardous	Progressive foam/absorber + support stroke + analysis against worst-case velocity
Common-cause EMI/lightning	Critical	CCA + physical/electrical segregation + zone separation
Common-cause fire	Critical	Fire zone segregation + barriers between survival-critical systems

16. PRINCIPAL SYSTEM INTERFACES

ID	Definition	Normal-State	Emergency Function	ICD
CAP-AF	Capsule ↔ airframe structural attachment	Transmit normal flight loads	Controlled release on Safety Kernel authorization	Doc 136
CAP-PAR	Capsule ↔ parachute system	Parachutes stowed; deployment inhibited	Deploy primary canopies after clearance verified	Doc 164
CAP-LG	Capsule ↔ eight landing supports	Ground loads during normal operations	Primary crash/landing load path after separation	Doc 156

ID	Definition	Normal-State	Emergency Function	ICD
CAP-AB	Capsule ↔ 16 airbags + sensors/inflators	Stowed; armed only under deterministic preconditions	Deploy under Safety Kernel authorization at impact	Doc 145
CAP-SEAT	Capsule floor ↔ 8 crashworthy seats + restraints	Normal seating load path	Stroke + energy absorption during impact	Doc 123
JET-PWR	Independent release power bus	Standby / trickle charge	Power to all release mechanisms on command	Doc 407
JET-CMD	Automatic/backup/manual command paths	Inhibited; monitoring only	Transmit release commands through independent paths	Doc 406
JET-SNS	Release-state sensing	Report attached state	Confirm released state at each stage	Doc 408
FUEL-ISO	Fuel isolation ↔ separation coordination	Normal fuel flow	Isolate fuel before or concurrent with separation	Doc 104
REC-LOC	Post-landing locator/communications	Dormant / periodic beacon check	Activate on post-separation; transmit position	Doc 405
AV-CTRL	Avionics/control ↔ Safety Kernel	Pass flight data for monitoring	Safety Kernel issues authorization; avionics executes	Doc 241

17. ENGINEERING / IP VALUE SURFACES

Rev20/22 concentrates on four primary claim candidates and retains remaining elements as supporting system narrative. Four narrowly-scoped, well-supported claims are stronger than nine broad assertions.

IP-A — Deterministic Safety Kernel Gating AI-Assisted Emergency Prediction. The specific combination of: (a) AI/ML-based anomaly prediction; (b) a hard deterministic precondition layer; and (c) selective, state-dependent (not all-or-nothing) actuation in a crewed rotorcraft survival system. Assessed as the strongest single claim in the package.

IP-B — Eight-Point Fixed Capsule-Integrated Landing Support with Asymmetric Failure State Management. The combination of: (a) eight-point, capsule-integrated, fixed, energy-absorbing mechanical support configuration; (b) individual support failure-state monitoring; and (c) state management adapting stability assessment to asymmetric failure configurations.

IP-C — State-Dependent Non-Rotor Selective Separation Sequencing for an Occupied Survival Capsule. Staged, state-dependent sequencing logic for selective release of non-rotor hazards from an occupied survival capsule, with explicit exclusion of the main rotor assembly and capsule-centric clearance verification gating recovery deployment. Distinguished from Ka-50/Ka-52 prior art.

IP-D — Integrated Multi-Layer Survival Chain for Crewed Rotorcraft. The specific combination of: (a) Deterministic Safety Kernel (IP-A); (b) state-dependent non-rotor selective separation (IP-C); (c) eight-point capsule-integrated impact attenuation with asymmetric failure management (IP-B); and (d) distributed canopy recovery — as a unified, interdependent system-architecture applied to a crewed rotorcraft survival capsule. System-level claims are harder to design around. Prior art (Ka-50: rotor separation + ejection seat; Soyuz: capsule + canopies) cannot anticipate the specific four-element combination. IP-D is assessed as the most commercially valuable claim in the package.

Supporting narrative (not standalone patent candidates): six-layer taxonomy; multi-mode recovery; partial/asymmetric-release management; immutable emergency event record; 16-airbag layout; crashworthy seat stroke specification. No novelty, patentability or freedom-to-operate conclusion is asserted by this baseline.

18. RISK REGISTER

Risk	Severity	Current Status	Priority
Stage S3 (capsule-to-airframe release) has no direct certified engineering analog	High	Open research item	P0 Critical
Low-altitude canopy deployment envelope may exclude most real accident profiles	High	Open — closure study required	P0 Critical
7,000 kg capsule mass not yet bottom-up verified	Medium	Planning allocation only	P0 Critical
No 6-DOF simulation evidence for any separation stage	High	Gate G4 not started	P0 Critical
Capsule clearance from rotor disk after separation not analyzed	High	Not yet addressed	P0 Critical
16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed	Critical	OI-18	P0 Critical
Airbag system mass 200 kg planning — detailed design may exceed budget	Medium	Corrected estimate	P0 High
Crashworthy seat design — stroke, load-distribution, combined-load cases not closed	Medium	[TARGET/STUDY]	P0 High
Six-layer independence against common-cause not demonstrated	High	CCA not performed	P0 Critical
No professional prior-art search for IP-A, IP-B, IP-C, IP-D	Medium	Prior-art posture only	P1 High
Certification basis not formally engaged with any regulator	Medium	Conceptual reference	P1 High
Main-rotor separation (deferred) remains structurally unanalyzed	Medium	Deferred research track	P2 Medium

19. PRIOR ART POSTURE

Class 1 — Ejection-seat rotorcraft (Ka-50/Ka-52): Rotor-blade-severing + crew ejection is directly relevant prior art for any main-rotor-related separation claim. This is why main-rotor separation is excluded from all primary IP claims (IP-A through IP-D).

Class 2 — Crewed spacecraft return capsules: Multi-canopy recovery (Soyuz 3 mains, Crew Dragon 3 mains), crashworthy seating, and independent post-landing life support are prior art for capsule-recovery and life-support elements. IP claims are narrowed to the rotorcraft-specific combination, not the general capsule concept.

Class 3 — Helicopter energy-absorbing landing gear: Energy-absorbing skids and struts exist in certified rotorcraft. IP-B novelty rests on the eight-point capsule-integrated configuration and individual support failure-state management — not on energy absorption in general.

Class 4 — Autonomous vehicle safety interlocks (IEC 61508, ISO 26262, DO-178C): Deterministic safety interlocks are well-developed. IP-A novelty is not in the concept of a safety interlock but in the specific combination of AI prediction + deterministic kernel + selective capsule-centric actuation in a crewed rotorcraft survival system.

Class 5 — Automotive and aviation airbag systems: Airbag systems are prior art. Any IP related to airbags would rest on the capsule-specific integration with the Safety Kernel, the sealed-volume environment, and the combined-attenuation architecture.

Class 6 — IP-D system-level combination: No single prior-art reference combines all four elements of IP-D (Safety Kernel + selective non-rotor separation + eight-point capsule attenuation + distributed canopy recovery) simultaneously in a crewed rotorcraft architecture. This is the basis for IP-D's strength. *A professional prior-art search against IP-A through IP-D is required before any patent filing or broad non-NDA disclosure.*

20. CRITICAL OPEN ITEMS REGISTER

ID	Open Item	Status	Pri	Closure Path
OI-01	Stage S3 capsule-to-airframe release has no certified engineering analog	OPEN — CRITICAL	P0	Mechanism design + 6-DOF + ground demo
OI-02	Minimum reliable canopy deployment altitude for 7,000 kg capsule not closed	OPEN — CRITICAL	P0	Sizing + inflation dynamics + envelope study
OI-03	7,000 kg capsule mass remains a planning allocation	OPEN — CRITICAL	P0	Component mass database + weighing plan
OI-04	No 6-DOF simulation for any separation stage	OPEN — CRITICAL	P0	Model hierarchy + correlation plan (G4)
OI-05	Partial and asymmetric separation detection and degraded recovery not detailed	OPEN — CRITICAL	P0	State sensing + FMEA + test cases
OI-06	Independence of six safety layers against common-cause not demonstrated	OPEN — CRITICAL	P0	CCA + fault injection + power/sensor diversity
OI-07	Capsule clearance from rotor disk after separation not analyzed	OPEN — CRITICAL	P0	6-DOF clearance analysis
OI-08	Occupant injury criteria and seat/restraint dynamic performance not closed	OPEN	P0	Dynamic seat tests + criteria selection
OI-09	Eight-support asymmetric failure and terrain interaction cases not closed	OPEN	P0	FEA + drop test matrix
OI-10	26,000 kg vehicle mass lacks component-level uncertainty tracking	OPEN	P1	Mass control plan + reserve policy
OI-11	Rotor/propulsion power-available vs power-required (OEI) not closed	OPEN	P1	Performance model + engine data
OI-12	Fuel isolation timing coordinated with separation not closed	OPEN	P1	Hazard analysis + sequence timing
OI-13	Professional prior-art search for IP-A through IP-D not performed	OPEN	P1	External IP counsel
OI-14	Certification basis not formally engaged	OPEN	P1	Regulator engagement plan
OI-15	ICDs for CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-* not issued	OPEN	P1	ICD set under configuration control
OI-16	16-airbag packaging, placement, mass, inflators, deployment logic not closed	OPEN	P0	Design + sled/impact tests
OI-17	Crashworthy seat stroke and load design (≥ 150 mm, $\leq 6,670$ N) not closed	OPEN	P0	Seat design + dynamic tests
OI-18	16-airbag deployment in sealed capsule — overpressure and gas-toxicity not analyzed	OPEN — CRITICAL	P0	Sealed-volume gas analysis + toxicity + venting/staging

21. PRELIMINARY PERFORMANCE ANALYSIS

21.1 Hover Power Study Calculation

Using actuator disk theory: $P_{ideal} = T^{3/2} / \sqrt{2\rho A}$ where $T = 26,000 \times 9.81 = 255,060$ N; $\rho = 1.225$ kg/m³ (ISA sea level); $A = 530.93$ m². **$P_{ideal} \approx 3.57$ MW [CALC]** — Study value; FM = 0.75 gives **$P_{shaft} \approx 4.76$ MW [CALC]**. Installed engine power must exceed P_{shaft} to account for profile power, transmission losses (~4–8%), tail rotor power (~8–12% of main rotor), accessory extraction, OEI margins, and hot/high performance. A plausible total installed power requirement is in the range of **6–8 MW**, corresponding to two turboshaft engines of approximately **3,000–4,000 kW each** — consistent with this mass class.

21.2 Mission Profile

Mission Parameter	Target Value	Status	Note
Design range	1,000 km	[TARGET]	Must be closed by mission fuel model
Crew + occupants	2 crew + 8 passengers	[TARGET]	8-person capsule limit
Cruise speed	TBD	[OPEN]	Depends on rotor/drag/engine selection
Cruise altitude	TBD	[OPEN]	Optimized for range/power balance
Takeoff fuel	TBD	[OPEN]	Included in 3,400 kg fuel budget
Cruise fuel	TBD	[OPEN]	Dominant range-driver
Reserve policy	TBD	[OPEN]	Per applicable civil/military requirement
IFR diversion reserve	TBD	[OPEN]	Typically 20–30 min additional fuel
Unusable fuel	~120 kg	[PLANNING]	Included in fuel budget
OEI diversion case	Required	[REQ]	Safety case requirement
Hot/high performance	Required	[REQ]	ISA +20°C at 3,000 m minimum target

The 1,000 km range at 26,000 kg MTOW with two turboshaft-class engines is achievable within the 3,400 kg fuel allocation if cruise SFC is in the range of 0.25–0.35 kg/(kW·h) — consistent with modern turboshaft-class engines — but requires closure of aerodynamic drag, rotor performance, and engine selection. No range claim is made until this closure is complete.

22. TECHNOLOGY READINESS LEVELS

TRL assessment per NASA/ESA TRL scale (1–9). Values are engineering judgments pending formal TRL review.

System / Subsystem	TRL Today	Basis for Assessment	Target G2	Target G8
Six-layer survival architecture concept	TRL 2	Architecture defined; no hardware	TRL 4	TRL 8
Deterministic Safety Kernel (IP-A)	TRL 2	Precondition set defined; no software	TRL 4	TRL 8
Capsule CFRP roll-cage concept	TRL 2–3	Similar CFRP capsule structures exist in spacecraft	TRL 5	TRL 8
Eight fixed landing supports	TRL 2	Similar energy-absorbers exist; capsule-integration novel	TRL 4	TRL 7
Stage S0–S2 (arming, fuel isolation)	TRL 3	Analogous systems exist in military jettison hardware	TRL 5	TRL 8
Stage S3 (capsule-to-airframe release)	TRL 1	No direct analog identified	TRL 3	TRL 7
3+2 canopy recovery	TRL 3	Three-main canopy architecture proven in spacecraft	TRL 5	TRL 7
Crashworthy seats (≥150 mm stroke)	TRL 3–4	Analogous seats exist in military helicopters	TRL 5	TRL 7
16-airbag capsule architecture	TRL 1–2	Airbag technology mature; capsule-integration novel	TRL 3	TRL 6
Sealed-capsule airbag gas management (OI-18)	TRL 1	No analogous system identified	TRL 3	TRL 6
Mass/CG/inertia model	TRL 2	Planning model only	TRL 5	TRL 8
6-DOF separation dynamics model	TRL 1	No simulation started	TRL 4	TRL 8
IP-A through IP-D patent claims	TRL 1	Architecture defined; no legal filing	Prior art searched	Filed

23. COMPETITIVE DIFFERENTIATION

Descriptive comparison, not a claim of superior certified safety performance. All HSH advantages are architectural concepts pending engineering verification.

Feature	Bell 412	AW139	S-92	Mi-26	HSH Rev22
Occupied survival capsule as final landing boundary	✗	✗	✗	✗	✓ [CONCEPT]
Deterministic Safety Kernel (AI-gated)	✗	✗	✗	✗	✓ [CONCEPT]
State-dependent selective non-rotor separation	✗	✗	✗	✗	✓ [CONCEPT]
Eight-point capsule-integrated impact attenuation	✗	✗	✗	✗	✓ [CONCEPT]
Distributed canopy recovery (3+2)	✗	✗	✗	✗	✓ [STUDY]
16-airbag occupant protection in capsule	✗	✗	✗	✗	✓ [TARGET/STUDY]
Six independently-capable survival layers	✗	✗	✗	✗	✓ [CONCEPT]
Independent post-separation life support	✗	✗	✗	✗	✓ [TARGET]
Explicit asymmetric/partial-failure state management	✗	✗	✗	✗	✓ [CONCEPT]
Crashworthy seats (≥150 mm stroke)	Some	Some	Some	Military	✓ [TARGET/STUDY]

Commercial positioning: HSH is not positioned as a replacement for existing helicopter models on cabin space or payload-fraction grounds. It is positioned as a distinct product category — a survivability-first rotorcraft — for operators whose primary requirement is occupant survival in catastrophic emergencies: offshore energy, search-and-rescue, medical evacuation, governmental/military VIP transport.

24. PRELIMINARY CAPSULE GEOMETRY ESTIMATE

Shows geometric plausibility of 8 occupants + airbags + crashworthy seats in a capsule of 7,000 kg. All values pending Document 112 closure.

Geometry Parameter	Preliminary Estimate	Status
Capsule outer diameter	≈ 3.2–3.6 m	[STUDY]
Capsule internal usable diameter	≈ 2.6–3.0 m (after roll-cage wall thickness)	[STUDY]
Capsule internal height (usable)	≈ 1.75–2.0 m (seated + headroom)	[STUDY]
Capsule outer shell thickness (CFRP + fairing)	≈ 100–150 mm per side	[STUDY]
Roll-cage member cross-section	≈ 60–100 mm diameter CFRP tubes	[STUDY]
Seat arrangement	2 rows × 4 seats, or circular/staggered 8-seat layout	[OPEN]
Seat pitch	≈ 650–750 mm	[STUDY]
Seat width	≈ 450–500 mm per position	[STUDY]
Seat stroke height occupied by absorber	≈ 150–200 mm below seat pan	[STUDY]

A circular capsule of ~3.4 m outer diameter, ~2.8 m internal diameter, ~1.85 m internal height accommodates 8 crashworthy seats in a 2×4 configuration with ≈700 mm pitch; 16 airbags stowed in headrests, sill-mounted and seat-back positions; 5-point restraints; emergency exits at minimum 2 locations with airbag-compatible egress paths; and internal volume for life support, emergency power, avionics and fire suppression. Geometric plausibility is supported at this preliminary level — bottom-up closure required (Documents 112, 113).

25. DETAILED AIRCRAFT MASS BREAKDOWN

Main Airframe — 5,000 kg: Fuselage primary load-bearing structure 1,450 kg; frames, bulkheads and longerons 650 kg; tail boom and tail structural assemblies 520 kg; engine and transmission support structure 380 kg; rotor mast/transmission support integration 260 kg; external skins, fairings and aerodynamic panels 310 kg; doors, access panels and hinges 190 kg; firewalls, barriers and structural fire protection 170 kg; structural fittings, lugs, fasteners and local reinforcements 260 kg; cabin/service floor structure outside capsule boundary 210 kg; integration margin remainder to 5,000 kg.

Main Rotor System — 2,100 kg: Main rotor blades (26.0 m system, 4–5 blades) 980 kg; rotor hub and blade retention/grips 280 kg; mast and rotor-head structural components 260 kg; swashplate and pitch-control hardware 170 kg; pitch links, control rods and bearings 110 kg; rotor vibration/dynamic-control hardware 80 kg; rotor lubrication/inspection/access 40 kg; configuration allowance + integration margin remainder.

Tail Rotor System — 480 kg: Tail rotor blades (4.0 m) 145 kg; hub and blade retention 65 kg; tail rotor gearbox 95 kg; drive shafting, couplings and bearings 60 kg; controls and pitch mechanism 35 kg; tail structure/interface hardware + margin remainder.

Engines and Transmission — 3,100 kg: Engine #1 complete installed 700 kg; Engine #2 complete installed 700 kg; engine accessories and engine-control hardware 180 kg; main gearbox / combining transmission 650 kg; intermediate/reduction transmission elements 180 kg; drive shafts and couplings 100 kg; lubrication, oil cooling and filtration 80 kg; engine/gearbox mounts and vibration isolation 60 kg; exhaust and hot-section installation 50 kg; propulsion integration margin remainder.

Fuel System + Usable Fuel — 3,400 kg: Usable fuel mass ~2,100 kg; fuel tanks (crash/fire-resistant structure) 280 kg; fuel pumps and boost/transfer units 90 kg; fuel lines, hoses, valves and fittings 110 kg; fuel filters, sensors and quantity measurement 55 kg; fuel isolation / emergency shutoff hardware 75 kg; venting, surge and inerting/fire-protection 70 kg; unusable / residual fuel allocation 120 kg; fuel system integration margin remainder.

Avionics / Electrical / Hydraulic / Thermal — 1,600 kg: Flight-control computers 180 kg; flight-control actuators/servos 210 kg; navigation/air-data/flight sensors 100 kg; communications/mission avionics 90 kg; electrical generation/distribution 190 kg; batteries/emergency electrical 90 kg; wiring/harnesses/connectors 170 kg; hydraulic pumps/reservoirs/lines/actuators 180 kg; thermal management/cooling 100 kg; environmental control/ventilation 70 kg; fire detection/warning/suppression interfaces 55 kg; health monitoring/data recording 45 kg; lighting/service/miscellaneous 20 kg; integration margin remainder. All values ASSUMPTION/STUDY pending bottom-up closure.

26. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

Gate	Evidence Objective	Key Evidence Items	Status
G0	Architecture Freeze	Safety philosophy; boundaries; interfaces; emergency states; airbag and seat concepts locked	Approaching closure
G1	Analytical Closure	Mass, CG, inertia, rotor, power, fuel, structural load paths, separation kinematics, seat stroke analysis	Not started
G2	Subsystem Demonstration	Release mechanisms, locks, absorbers, parachute, airbag module, seat stroke, sensors, independent power	Not started
G3	Integrated Ground Demonstration	End-to-end sequence; timing; state sensing; capsule clearance; airbag deployment in representative volume	Not started
G4	6-DOF Simulation	Vehicle and capsule dynamics; partial/asymmetric release; parachute inflation; sealed-capsule airbag gas	Not started
G5	Structural / Crash Testing	Capsule, attachments, seats (dynamic stroke test per FAR 27.562 proxy), airbag sled tests, supports, energy absorbers	Not started
G6	Recovery System Testing	Canopy deployment; inflation loads; failure cases; stability; landing performance	Not started
G7	Flight Development	Progressive envelope expansion; normal and emergency recovery validated in flight	Not started

Gate	Evidence Objective	Key Evidence Items	Status
G8	Certification / Qualification	Full evidence package against selected certification basis	Not started

Terminology: Specified = defined in design; Analyzed = documented analytical result; Simulated = supported by a model; Tested = demonstrated in a defined test; Verified = evidence shows compliance; Validated = suitability established for intended use; Qualified/Certified = formal processes complete. **As of Rev22 FULL: all HSH-00 claims are Specified or Studied. None are Analyzed, Simulated, Tested, Verified, Validated, Qualified, or Certified.**

27. ENGINEERING MATURITY FRAMEWORK

Configuration Item	Rev 22 Control Rule	Closure Evidence Required
MTOW / design mass	Freeze only after bottom-up mass model and reserve policy reconciled	Mass report + uncertainty analysis
Rotor geometry	Maintain 26.0 m target until performance study closes	Rotor sizing + power model
Capsule mass	7,000 kg planning target; no closure claim	Bottom-up mass budget (Doc 113)
Occupancy	8 persons maximum — LOCKED for Generation 1	Configuration control only
Recovery	3 primary + 2 reserve study architecture	Sizing + inflation dynamics + failure testing
Separation	Non-rotor candidate set only in core baseline	6-DOF + clearance + ground demonstration
Safety authorization	Deterministic kernel remains authoritative	Requirements + software/hardware verification
Airbags	16 total (2 per occupant) TARGET/STUDY	Packaging + sled/impact tests + OI-18 gas analysis
Crashworthy seat stroke	≥150 mm stroke; ≤6,670 N criterion	Dynamic seat tests + analysis

Maturity roadmap: Architecture control (G0) → Analytical closure (G1) → Subsystem design (G2 prep) → Ground demonstration (G3) → Model validation (G4) → Structural/recovery tests (G5/G6) → Flight development (G7) → Qualification/certification (G8).

28. MASTER CONFIGURATION RECORD

Configuration Item	Rev 22 Baseline	Status
Aircraft programme name	HSH Heavy Safe Helicopter	LOCKED
Founder	Amirbahador Atai	CONTROLLED
Generation	Generation 1	LOCKED
Document number	HSH-00	LOCKED
Revision	Rev 22 FULL	BASELINE
Capsule occupancy maximum	8 persons	LOCKED LIMIT
Vehicle design mass	26,000 kg	TARGET / OPEN
Capsule mass allocation	7,000 kg	PLANNING / OPEN
Main rotor target	26.0 m	TARGET / OPEN
Tail rotor target	4.0 m	TARGET / OPEN
Range target	1,000 km	TARGET / OPEN
Landing supports	8 fixed energy-absorbing	CONCEPT / OPEN
Recovery	3 primary + 2 reserve canopies	STUDY / OPEN
Occupant airbags	16 (2 per occupant)	TARGET / STUDY
Crashworthy seat stroke	≥150 mm; ≤6,670 N	TARGET / STUDY
Core separation	Selective, staged, non-rotor only	CONCEPT / OPEN
Main-rotor separation	Excluded from core safety case	DEFERRED RESEARCH
Safety authorization	Deterministic Safety Kernel	CONCEPT / OPEN
IP claims	IP-A, IP-B, IP-C, IP-D	CONCEPTUAL / NOT FILED
Development gate	G0 approaching closure	CURRENT

29. ENGINEERING BASIS AND EXTERNAL REFERENCES

Predecessor documents: LSH-00 Rev3.2 (six-layer philosophy; independent initiation; controlled separation; eight supports; capsule-as-final-boundary concept); USCC Doc 00 Rev1.3 (jettison sequence; deterministic safety authorization); HSC-00/HSH-00 Rev12–21 (progressive engineering development and correction).

External engineering references: EASA CS-29 Amendment 12: Large rotorcraft certification standard; jettisonable configurations require clear trajectory, no damaging interaction, acceptable post-jettison stability and performance. FAR 27.562 / CS-27 Dynamic seat test requirements: Proxy reference criterion for ≤6,670 N peak spinal compressive load used in crashworthy seat specification (§10B). MIL-STD-1290A: Helicopter crashworthiness design criteria; relevant to capsule structural design, seat design, landing-support design and occupant injury criteria. NASA TRL scale: Used for TRL assessment in §22.

30. DESIGN PHILOSOPHY — CONTROLLED SURVIVAL, NOT A SINGLE DEVICE

HSH Generation 1 integrates multiple survival mechanisms into one state-aware architecture. The vehicle does not depend solely on a parachute, solely on autorotation, solely on a detachable cabin, or solely on automatic software. Removing main-rotor

separation from the core safety case — introduced in Rev13 and retained through Rev22 — **strengthens** rather than weakens the layered proposition: a safety case that requires its riskiest, least-proven capability to succeed is not actually layered. Rev20/22 extends the impact-attenuation layer with 16 airbags and crashworthy seats specified to the correct engineering standard — these are complementary protections, not substitutes for structural crashworthiness, landing supports, or recovery canopies. The strongest future version of HSH Generation 1 is the version in which every important claim has a visible path from requirement to model to test to controlled evidence. Documentation is not a substitute for evidence. The path from this baseline to G8 is long and requires real engineering work — but the architecture is now defined clearly enough, honestly enough, and with sufficient IP specificity to serve as a credible starting point for that work.

31. FINAL GENERATION 1 BASELINE STATEMENT

HSH-00 Rev22 FULL is the definitive Generation 1 conceptual master baseline for the HSH Heavy Safe Helicopter. It is the document a technical due-diligence team from a potential IP licensee should receive as the starting point for evaluation.

What is defined: the full six-layer survival architecture; the Safety Kernel and its precondition set; the staged S0–S5 separation timeline with honest confidence-basis assessment; the 3+2 canopy recovery architecture with preliminary sizing; the 8-point capsule-integrated landing-support system; the 16-airbag and crashworthy-seat occupant-protection architecture (with corrected engineering specifications); four IP claims (IP-A through IP-D) with prior-art awareness; a full open-items register (OI-01 through OI-18); a TRL assessment; a competitive differentiation table; a preliminary capsule geometry check; a mission profile framework; and the diligence enhancement layer §§32–41.

What remains to be proven: everything in the open-items register. The gap between architecture and evidence is the scope of the Generation 1 engineering document tree (Documents 001–500).

ENHANCEMENT LAYER — SECTIONS 32 TO 41

32. ENGINEERING CLAIM ARCHITECTURE

Purpose: convert the principal HSH engineering and IP propositions into traceable engineering claims. Each claim is connected to the evidence that must exist before the claim can be treated as demonstrated. This is an architecture for closure, not a statement that the claims are already proven.

Claim ID	Engineering Claim	Primary Evidence Required	Verification Path	Tree Docs	Current Basis
EC-001	Occupied survival capsule can become the final landing boundary in the defined emergency architecture	Requirements, structural load paths, attachment analysis, separation dynamics, clearance evidence	Analysis → 6-DOF simulation → ground demonstration → applicable testing	REQ / SEP / DYN / VER	Specified / Studied
EC-002	State-dependent selective separation can release designated non-rotor hazards while preserving capsule survivability	Release mechanism design, authorization logic, partial/asymmetric cases, failed-release fallback	Analysis → mechanism test → integrated ground demonstration	SEP-017 to SEP-050	Specified / Open
EC-003	Deterministic Safety Kernel can remain the authoritative authorization boundary for safety-critical actuation	Requirements, precondition logic, independence analysis, formal/robustness evidence	Requirements verification → formal methods / fault injection → qualification evidence	SAF-002 to SAF-050	Specified / Open
EC-004	Eight capsule-integrated landing supports can provide progressive impact attenuation with asymmetric failure management	Load cases, FEA, absorber characteristics, terrain cases, failed-support redistribution	Analysis → drop testing → integrated crashworthiness evidence	Landing-support / crash series	Specified / Open
EC-005	3 primary + 2 reserve canopy architecture can recover the 7,000 kg capsule within a validated deployment envelope	Sizing, inflation dynamics, failure tolerance, minimum altitude envelope	Sizing analysis → deployment tests → envelope closure	REC / PAR series	Study / Open
EC-006	16-airbag + ≥150 mm stroke seat architecture can limit occupant loads under defined impact conditions without defeating egress or sealed-capsule environment	Packaging, deployment logic, sled/impact tests, sealed-volume gas analysis (OI-18)	Design → sled/impact → gas analysis → combined attenuation evidence	CAP-AB / CAP-SEAT / OI-18	Target / Open
EC-007	IP-A through IP-D remain coherent conceptual claim surfaces supported by the engineering architecture	Prior-art posture, claim definitions, engineering traceability	Professional prior-art search → claim drafting → legal review	IP domain docs	Conceptual / Not filed

33. IP-TO-ENGINEERING TRACEABILITY MATRIX

This matrix connects the four principal IP surfaces to the engineering domains that must close before any claim can be treated as more than conceptual.

IP ID	Claim Surface	Engineering Domains That Must Close	Primary Tree Anchors	Critical Dependencies
IP-A	Deterministic Safety Kernel gating AI-assisted emergency prediction	Safety Kernel requirements; precondition set; independence from AI/ML; fault injection; software/hardware assurance	SAF / AV / VER series	OI independence + formal verification
IP-B	Eight-point fixed capsule-integrated landing support with asymmetric failure state management	Support geometry; energy absorbers; failed-leg FEA; drop tests; terrain cases	Landing-support / crash / STRUCT	OI-09 + G5 evidence

IP ID	Claim Surface	Engineering Domains That Must Close	Primary Tree Anchors	Critical Dependencies
IP-C	State-dependent non-rotor selective separation sequencing for occupied survival capsule	Candidate component list; S0–S5; release mechanisms; clearance verification; 6-DOF	SEP / DYN series	OI-01, OI-04, OI-05, OI-07
IP-D	Integrated multi-layer survival chain (Kernel + selective separation + eight-point attenuation + distributed canopy recovery)	All of IP-A/B/C evidence plus recovery envelope and combined system demonstration	Cross-domain G3–G6	System-level integration gates

34. BUYER DUE-DILIGENCE READINESS MATRIX

Intended to let an industrial acquirer review the baseline without reconstructing the entire programme architecture from 500 documents. Status follows Rev20 terminology.

Due-Diligence Question	HSH-00 Answer	Primary Supporting Documents	Closure State
What is the core system proposition?	Six-layer survival architecture centered on an occupied survival capsule, deterministic Safety Kernel, selective non-rotor separation, 8-point impact attenuation and 3+2 recovery	HSH-00 §§2, 6, 12, 30	Defined
What is technically defined today?	Top-level geometry, mass allocations, occupancy lock, Safety Kernel preconditions, S0–S5 concept, airbag/seat architecture, IP-A–D posture, OI-01–OI-18	HSH-00 §§2–5, 7–17, 21–31	Defined / Studied
What is not yet proven?	The open-item register remains the evidence gap, especially S3 release, low-altitude recovery, mass bottom-up, 6-DOF, OI-18 gas analysis	HSH-00 §§18–20	Open
What is the evidence path?	G0 architecture freeze through G8 certification / qualification; E0–E5 evidence ladder in §37	HSH-00 §26; VER series	Defined / Not started
What are the principal residual risks?	S3 release, low-altitude recovery envelope, 7,000 kg mass closure, sealed-capsule airbag environment, six-layer independence	HSH-00 §§18–20	Open / Critical
What is the IP posture?	IP-A through IP-D are candidate conceptual surfaces with prior-art awareness; not filed; professional search required	HSH-00 §§17, 19	Conceptual / Legal open
What configuration is locked?	Generation 1 maximum occupancy of 8 persons; no main-rotor separation in core safety case	HSH-00 §§1, 2, 28	Controlled

35. SAFETY CASE ARCHITECTURE

The safety case is organized as a layered argument: the vehicle shall provide multiple independent means of protecting occupants, ordered from conventional recovery to capsule separation and post-landing survival. The safety case does not rely on main-rotor separation. Each layer requires defined requirements, hazards, verification methods and residual-risk acceptance.

Safety-Case Block	Argument	Evidence Required	Principal Residual Risk
Layer L1–L3 initiation	Automatic primary, automatic backup, and non-bypassable crew authority	Independence analysis, fault injection, HMI verification	Common-cause defeat of all initiation paths
Layer L4 selective separation	State-dependent release of designated non-rotor hazards only	Mechanism design, S0–S5 timing, 6-DOF, ground demonstration	Unintended release; incomplete/asymmetric release
Layer L5 direct-impact survival	Capsule structure, roll-cage, 8 supports, crashworthy seats, 16 airbags	FEA, drop/sled tests, OI-18 gas analysis, injury criteria	Bottom-out; overpressure; egress blockage
Layer L6 recovery / post-event	3+2 canopies, independent power, locator, life support	Sizing, deployment tests, minimum altitude envelope	Low-altitude non-availability; canopy failure
Safety Kernel boundary	Deterministic authorization of safety-critical actuation	Requirements, formal/robust verification, independence from AI	AI bypass; incomplete preconditions

36. FAILURE CONTAINMENT PHILOSOPHY

The architecture identifies unintended separation, partial/asymmetric release, sensor loss, power-bus failure, AI/ML failure, fire during separation, released-component collision, parachute out-of-envelope deployment, landing-support failure, airbag inadvertent/non-deployment, sealed-capsule overpressure/toxicity (OI-18), and seat stroke bottom-out. Containment philosophy: prevent common-cause defeat of multiple layers; detect degraded states; define safe fallback for every failed-release configuration; preserve human authority; keep AI advisory at the safety boundary.

Failure Class	Containment Design Response	Program Rule
Unintended separation	Mechanical locks + electrical inhibits + command validation + monitoring	Quantitative safety objective + CCA
Partial / asymmetric release	State sensing + asymmetric-load analysis + degraded recovery mode	First-class emergency state, not an afterthought
Sensor loss / bad data	Validity monitoring + diverse sensing + safe-state logic	Kernel must fail safe
Power-bus failure	Independent survival power for critical functions	No single bus defeats L1–L6 critical path
AI/ML failure	Deterministic Safety Kernel remains authoritative	Non-negotiable certification boundary

Failure Class	Containment Design Response	Program Rule
Sealed-capsule airbag gas (OI-18)	Sealed-volume analysis + staged/vented deployment options	Must close before deployment logic freeze

37. EVIDENCE LADDER

All HSH-00 claims are Specified or Studied. The ladder is the controlled progression from specification to demonstrated evidence.

Ladder Step	Meaning	Minimum Evidence Content	Typical Gate
E0	Specified / Studied	Requirement or concept written with status label	HSH-00 (current)
E1	Analyzed	Documented analytical result with stated assumptions and uncertainty	G1 mass, loads, kinematics
E2	Simulated	Model result with stated fidelity and validation status	G4 6-DOF; OI-18 gas model
E3	Subsystem tested	Defined test article, conditions, instrumentation, results	G2 mechanisms, seats, airbags
E4	Integrated demonstrated	End-to-end ground sequence under controlled conditions	G3 integrated demonstration
E5	Flight / qualification evidence	Flight development and formal qualification against basis	G7–G8

38. SUBSYSTEM TRL + EVIDENCE + GATE MATRIX

Extends the TRL table by adding the evidence needed to move each subsystem toward programme gates. TRL values are not re-scored upward by documentation alone.

Subsystem	Rev20 TRL	Evidence Needed Next	Primary Gate	Target Evidence
Six-layer survival architecture	2	Independence analysis, CCA, fault injection	G1–G3	E1–E4
Deterministic Safety Kernel (IP-A)	2	Requirements, formal/robust verification, independence from AI	G1–G2	E1–E3
Capsule CFRP roll-cage	2–3	FEA, joint/node design, crash load cases	G5	E1–E3
Eight fixed landing supports	2	Absorber characterization, drop tests, asymmetric cases	G5	E2–E4
Stage S0–S2 (arming, fuel isolation)	3	Mechanism tests, timing, isolation verification	G2–G3	E3–E4
Stage S3 (capsule-to-airframe release)	1	Mechanism design, 6-DOF, ground demonstration	G2–G4	E1–E4
3+2 canopy recovery	3	Sizing, inflation, failure cases, min altitude	G6	E2–E4
Crashworthy seats ≥ 150 mm / $\leq 6,670$ N	3–4	Dynamic seat tests (FAR 27.562 proxy)	G2–G5	E3–E4
16-airbag capsule architecture	1–2	Packaging, deployment, sealed-volume gas, sled tests	G2–G5	E2–E5
Mass / CG / inertia model	2	Bottom-up mass database, uncertainty, CG/inertia closure	G1	E2
6-DOF separation dynamics	1	Model specification, correlation plan	G4	E3
IP-A through IP-D	1	Prior-art search, invention disclosures, claim drafting, legal review	External IP gates	E0–E1 legal

39. MASS CLOSURE CONFIDENCE MATRIX

Controlled 26,000 kg accounting model and 7,000 kg capsule allocation with a closure-confidence view so each major mass block has a defined route from allocation to evidence.

Mass Block	Rev20 Planning	Current Basis	Closure Evidence	Tree Anchor	Impact
Complete survival capsule	7,000 kg	Planning allocation	Bottom-up component database + weighed articles	MASS-003; capsule	High
Main airframe	5,000 kg	Assumption / study	Bottom-up structural and systems mass model	MASS-004	High
Main rotor / hub / controls	2,100 kg	Assumption / study	Rotor component mass model + design definition	MASS-005; ROT	High
Tail rotor / gearbox / drive	480 kg	Assumption / study	Component definition + supplier / design data	ROT / PROP	Medium
Engines / transmission / mounts	3,100 kg	Assumption / study	Selected engine and gearbox data + installation model	MASS-006; PROP	High
Fuel system + usable fuel	3,400 kg	Assumption / study	Fuel accounting + tank/pump/line definition	MASS-012; FUEL	High
Rapid separation system	1,000 kg	Assumption / study	Mechanism and interface mass model	SEP domain	High
Avionics / electrical / hydraulic / thermal	1,600 kg	Assumption / study	Subsystem mass models + harness estimate	AV / ELEC	Medium
Operating fluids / consumables	320 kg	Assumption / study	Fluid accounting	MASS series	Low
Growth / configuration reserve	2,000 kg	Program reserve	Reserve policy + uncertainty tracking	Mass control plan	High

Mass Block	Rev20 Planning	Current Basis	Closure Evidence	Tree Anchor	Impact
Airbags (inside capsule 200 kg line)	200 kg	Corrected estimate	Module + ECU + harness + mounting bill of materials	CAP-AB docs	High
Crashworthy seats + restraints	700 kg	Planning line	Seat supplier data + stroke mechanism mass	CAP-SEAT docs	High

40. CONFIGURATION BASELINE DIAGRAM (TEXTUAL)

Textual configuration map of the locked and target elements. Visual CAD is out of scope for this document.

Configuration Class	Content	Control Rule
LOCKED LIMIT	8 persons maximum occupancy	Any change → full re-analysis
LOCKED CONCEPT	No retractable landing gear; capsule central; main rotor not in core separation set	Impact assessment required to alter
TARGET geometry	26.0 m main rotor; 4.0 m tail rotor; 26,000 kg design mass; 7,000 kg capsule	Bottom-up and performance closure
TARGET survival hardware	8 fixed energy-absorbing supports; 16 airbags; ≥150 mm stroke seats ≤6,670 N; 3+2 canopies	Design + test evidence
NORMAL FLIGHT LOAD PATHS	Airframe ↔ CAP-AF ↔ capsule structure ↔ seats	Primary flight loads
EMERGENCY LOAD PATH / FINAL LANDING BOUNDARY	Released capsule ↔ 8 supports ↔ crashworthy seats ↔ airbags ↔ occupants	Post-separation impact
CONTROLLED INTERFACES	CAP-AF, CAP-PAR, CAP-LG, CAP-AB, CAP-SEAT, JET-*, FUEL-ISO	ICD set under configuration control
STATE-DEPENDENT TRANSITION	Normal → Anomaly → Confirm → Assess → Recover if possible → Authorize → Selective separation → Clearance → Recovery → Landing	Emergency state machine
INDEPENDENCE PRINCIPLE	Physical, electrical and software segregation where required; human authority preserved; AI advisory only at the safety boundary	Non-negotiable

Configuration-control anchor: any change to occupancy, mass, rotor geometry, separation architecture, recovery architecture, landing-support count/geometry, airbag count/placement, seat stroke or Safety Kernel preconditions triggers the impact-assessment process already defined in this baseline.

41. DOCUMENT DEPENDENCY / PARENT–CHILD ARCHITECTURE

The Master Engineering Document Tree contains requirements, subsystem design, safety, dynamics, verification, certification, IP and programme-management documents. This section adds a dependency discipline so that each lower-level document can identify its controlling parent, upstream inputs, downstream verification artifact and closure gate.

Document Layer	Example Parent	Example Child / Output	Dependency Rule
HSH-00 Master Baseline	HSH-00 Rev22 FULL	REQ / SAF / MASS / ROT / SEP / REC / AV / DYN / VER / IP domains	Master baseline controls architecture, terminology, locked limits and gate definitions
System Requirements	REQ-001–040	Subsystem requirements, derived requirements, RVM	Child requirements shall trace upward to parent REQ and to HSH-00
Safety domain	SAF plans + FHA/PSSA/SSA	CCA, PRA, ZSA, Safety Kernel verification	No safety claim without upward trace to HSH-00 layers and OI register
Capsule domain	CAP requirements + design	Roll-cage, seats, airbags, supports, mass, tests	Must respect 8-person lock, 7,000 kg allocation, OI-16/17/18
Separation domain	SEP requirements + design	Mechanisms, S0–S5, 6-DOF, ground demo	Main rotor remains excluded from core candidate set
Recovery domain	REC / PAR	3+2 sizing, deployment, envelope	OI-02 minimum altitude is critical path
Verification domain	VER plans + matrices	G0–G8 evidence packages	Evidence ladder E0–E5 applies
IP domain	IP-A/B/C/D descriptions	Prior-art search, claim drafts, FTO	No novelty assertion without external legal work

END OF HSH-00 REVISION 22 FULL

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